

LOGIC SIM v1.20.7

PINAGSAMANG REQUIREMENTS PARA KAY CLAUDE

Grouped version · Mas kaunting duplicate · Plain Tagalog/Taglish

Layunin: Makumpleto at ma-verify ang existing simulation functionality nang mas mabilis. Pinagsama ang magkakaparehong requirements. Pinanatili ang lahat ng 20 original findings, pero inilagay sila sa tamang group para hindi paulit-ulit ang instructions.

Version note: Ituring na v1.20.7 ang app at findings report. Ang v1.27.0 sa dating report header ay typo lamang.

Terminology: Ang values sa Simulation Mode ay simulation values. Gamitin ang “input value,” “converted/calculated value,” at “output value.” Huwag tawaging actual/live plant values.

1. Priority at scope

Priority	Requirement
P1 — Simulation correctness	Ayusin ang LINEAR FX scaling, signal mapping, block calculations, at cross-sheet continuity.
P2 — Analog behavior	I-verify ang COS, MAN, math, NOT, transmitter selection, RATE/ramp, bypass, at actuator response.
P3 — Verification	Gumamit ng reusable tests, expected-versus-actual results, evidence, at retest status.
P4 — Optional improvements	Side-by-side sheets at cosmetic UI changes. Huwag unahin habang may core simulation issues.

- Unahin ang pagkumpleto ng existing simulation; huwag magdagdag ng unrelated features.
- Suriin ang current source bago mag-edit. Gumawa ng focused changes at iwasan ang broad refactor.
- Ayusin ang shared/root cause kaysa maglagay ng magkakahiwalay na patch sa bawat sheet.
- Huwag sabihing fixed o fully tested kung walang test evidence.

2. Group A — LINEAR FX, values, ranges, at units

- Gamitin ang tamang built-in LINEAR table, input/output range, units, at conversion direction.
- Supportahan ang engineering units → percent, percent → engineering units, at non-percent → non-percent kapag defined ng table.
- Huwag ipagpalagay na percent ay laging 0–1 o 0–100. Sundin ang representation sa app, drawing, at table.
- Panatilihin magkahiwalay ang input value, converted/calculated value, at output value. Halimbawa: 10,000 cal input ay puwedeng manatiling displayed bilang 10,000 cal habang 50% ang converted FX value.
- Kung ang table ay 8,000 cal = 0% at 12,000 cal = 100%, ang 10,000 cal ay dapat maging 50%. Example lang ito; actual drawing/table ang authority.
- Kapag missing o ambiguous ang table, mag-warning at markahang NEEDS REVIEW. Huwag tahimik na gumamit ng $y=x$ /pass-through at magmukhang valid ang result.
- I-apply lang ang range limit sa signal/block kung saan ito defined. Huwag generic na i-clamp ang lahat ng intermediate values.
- Sa COS, i-check ang compatibility ng SV units/range. Sa MAN, ihiwalay ang manual input/setpoint range sa output/MV range; hindi automatic na pareho ang mga ito.

Required test	Expected check
Low / midpoint / high	Tama ang conversion ayon sa LINEAR table.
Percent ↔ engineering units	Tama ang conversion sa parehong direction.
Out-of-range input	Defined behavior ang masunod; huwag mag-clamp maliban kung required.
Missing/ambiguous table	Warning + NEEDS REVIEW; walang silent pass-through.

3. Group B — Signal connections, symbols, at sheet links

- Ang valid wire geometry/junction/pin mapping ang basehan ng connection. Hindi sapat ang magkalapit na tag o symbol.
- Ang tags/addresses ay identifiers; hindi sila automatic na source ng value.
- Lahat ng labels/wire values sa iisang valid net ay dapat consistent ang simulation value.
- Kung unknown ang symbol, missing ang pin, o unresolved ang connection, mag-warning at huwag ituring na verified ang output.

- I-verify ang FROM DITL / ABC references at cross-sheet continuation. Panatilihin ang signal path nang hindi nag-iimbento ng connection, ramp, o delay.
- Test item: high/low selector at active-wire highlighting; selected path at visual highlight ay parehong kailangang tama.
- Test item: math block operation at input pin mapping gamit ang known values; NOT gate gamit ang truth table at traced inputs.

4. Group C — Transmitter bad signal at SIG.AB

- Magkaroon ng simulation-only controls sa supported transmitter inputs: Normal, Bad Signal, Manual Test Value, Restore Normal.
- Ang Bad Signal ay dapat hiwalay na signal-quality/status condition kung suportado ng logic model; hindi ito simpleng numeric zero.
- Ang zero ay maaaring valid measurement. Huwag ituring na failed transmitter dahil zero lang maliban kung malinaw iyon sa drawing.
- Ipakita ang forced/bad status sa diagram at trace, pati ang downstream effect. Restore dapat magtanggag ng temporary injection nang hindi binabago ang source drawing.
- Walang writes o connection sa actual plant DCS.

SIG.AB case	Primary TX	Secondary TX	I-verify
1	Normal	Normal	Normal selection/output.
2	Bad Signal	Normal	Fallback/exclusion ayon sa drawing.
3	Normal	Bad Signal	Primary path at epekto ng secondary failure.
4	Bad Signal	Bad Signal	Defined failure response; huwag manghula.
5	Normal = 0	Normal, nonzero	Zero ay hindi automatic na bad.

5. Group D — COS/T-switch, RATE/ramp, bypass, final element

- I-test ang COS at T-switch nang magkahiwalay at kapag magkalapit. Dapat valid connection/mapping, hindi proximity lang, ang basehan ng association.
- I-test ang ramp-limited path at bypass path nang magkahiwalay gamit ang parehong input change.
- Kapag active ang bypass, sundin ang actual logic sa drawing. Kung may ramp pa rin, alamin kung may downstream ramp block o parsing error.
- Kapag hindi bypassed, i-verify ang RATE value at units. Ang missing/zero rate ay hindi dapat tahimik na magdulot ng unintended instant step.
- Ihiwalay ang command sa actuator position: maaaring instant ang digital OPEN/CLOSE command, habang gradual ang MOV/control-valve position kung may defined stroke time/model.
- Huwag magdagdag ng ramp/delay na wala sa drawing. Ang cross-sheet connection ay hindi automatic na nagdadagdag ng delay.
- Grey analog signals: maaaring expected sa View mode. Kung grey pa rin sa Simulation mode kahit active ang signal, i-reproduce at i-test.

6. Group E — Test workflow para mas mabilis matapos

Hindi ito performance optimization. Ang goal ay mapabilis ang pagkumpleto ng simulation sa pamamagitan ng mas maayos na debugging, reusable tests, at mas kaunting rework.

- Step 1: Suriin ang source at tukuyin ang shared LINEAR, mapping, at block-calculation logic.
- Step 2: Pumili ng isang representative ABC sheet; ayusin at i-verify muna ang LINEAR FX at signal mapping.
- Step 3: Gumawa ng reusable test cases na may verified expected values.
- Step 4: Pagkatapos ng bawat fix, i-retest ang affected case at relevant working functions, kasama ang digital simulation.
- Step 5: Expand sheet by sheet. Huwag mag-claim ng full coverage kung hindi lahat ng relevant cases ay na-test.
- Huwag munang gumawa ng malaking independent Test Bench. Minimal presets na gumagamit ng existing simulator, trace, at audit components ang unahin.

Test result labels: PASS = tugma sa verified expected result; FAIL = hindi tugma; NEEDS REVIEW = hindi malinaw ang expected behavior/reference; NOT TESTED = walang test evidence.

Bawat test record: sheet, tag/block, input conditions, expected value/unit, actual value/unit, source ng expected result, trace/evidence, status, correction, at retest status.

7. Lahat ng 20 original findings — grouped index

Pinanatili ang lahat ng original findings. Ang detalyadong instructions ay nasa Groups A–F sa itaas; huwag gumawa ng duplicate ticket kung covered na ng parehong root cause.

ID	Original finding	Group
01	High/Low selector at active-wire highlighting	B
02	Forced wire/net value label	C / E
03	Two sheets side-by-side	Optional
04	Manual input range enforcement	A
05	COS range	A / D
06	High/Low selector transition behavior	B / D
07	Unknown/missing symbols or pins	B
08	Ramp path vs bypass path	D
09	Built-in LINEAR FX tables	A
10	Percent scaling in LINEAR FX	A
11	ABC-002 32% minimum airflow constant	F
12	Math block operation	B
13	Grey analog signals	D
14	NOT gate	B
15	Final element behavior across sheets	B / D
16	SIG.AB selection circuit	C
17	Signal ranges / clamping	A
18	FROM DITL / ABC references	B
19	RATE setting and units	D
20	T-switch near COS	D

8. Additional verification item

ABC-002 32% minimum-airflow constant: i-compare sa actual drawing/reference. Huwag ipagpalagay na mali agad. Itala ang source at expected value bago mag-test.

9. UI scope sa batch na ito

- Alisin muna sa UI ang Plant Model feature at Plant simulator/window ayon sa naunang scope.
- Temporarily alisin ang Save, Save As, Open Project, at Import toolbar actions. Huwag itong ihalo sa audit report save/export.
- Panatilihin ang dark theme preference at built-in LINEAR tables.
- Side-by-side sheets at cosmetic improvements ay optional; huwag unahin sa core simulation.

10. Ano ang kailangang ibalik ni Claude

- Confirmed bugs, unconfirmed symptoms, at NEEDS REVIEW items.
- Files/functions na binago at maikling paliwanag ng behavior change.
- Tests na talagang pinatakbo, expected vs actual, at evidence.
- Regression results para sa affected sheet at relevant working functions.
- Remaining issues/risks. Huwag mag-claim ng full validation kung hindi iyon nasubukan.

Acceptance goal: tama ang LINEAR conversion, signal mapping, at block behavior; malinaw ang unresolved errors; may evidence ang tests; at mas mabilis nang ma-verify ang bawat ABC sheet nang hindi paulit-ulit ang parehong trabaho.

END — Mandatory core fixes first. Optional UI improvements later.